

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

What to Do with Defects in Software Development



Learning Outcome:

Describe ways of solving problems and getting things done in written and spoken situations and in work and social contexts.

Train the system users on application structuring and management, according to the established plan and user profile, according to the organization policies.

Can Do Statements:

- I can simply explain the concept of defect tracking.
- I can talk about the purpose of defect tracking.
- I can discuss the risks of not tracking defects for a software development organization.

Ask your partner:

- How do you think a software development team should treat defects? Should they enter all defects in a database? Should they just communicate them to the person responsible for the defect?
- Why do you think a software development project constantly gets defects?





Let's Explore!

Why do some organizations consider defect tracking so important?



Instructions: Read the text below.

Arnold: I'm still getting acquainted with software development terms and I'm not sure what defect tracking is for.

Carl: Defect tracking is the process in which logged defects are tracked from the start to the end of a project. If we hadn't tracked the defects for this project, we wouldn't have addressed them on time or, even worse, we wouldn't have fixed them at all.

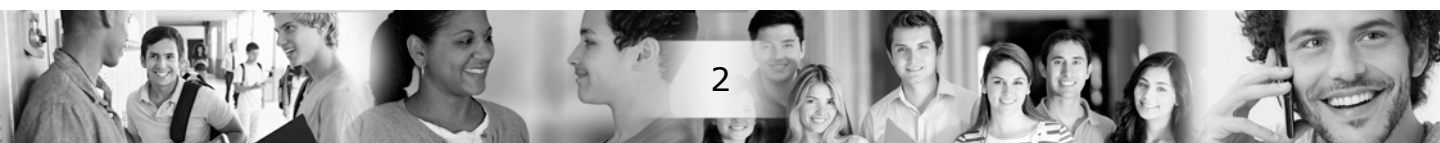
Arnold: As far as I'm concerned, many organizations consider this to be an indispensable process for the development of their projects. What makes it so important?

Carl: On the course of a project, many defects are going to occur as an undeniable human error. Once these defects have been detected, they need to be tracked, usually, by using a defect tracking system in which the details about defects are entered. In this way, the information is going to be available for the other team members. Sometimes, these defects need to be validated, they're reviewed by different people and then finally, they arrive at the person responsible for fixing it. That's why it's utterly important to track defects, this is to ensure that there's clear information about how the defect should be treated.

Arnold: So, most of the defects wouldn't have been fixed if they hadn't been tracked?

Carl: Most likely. Moreover, tracking defects helps to find defect patterns that point out where most defects come from. For instance, if we hadn't looked at these patterns, we wouldn't have realized that we needed to improve the requirements' definition and coding areas.

Yet another advantage of defect tracking is the improvement in models for testing. If we hadn't compared the defects found in the testing with those found in production, we wouldn't have realized how many defects our testing processes we are not detecting.





Let's Practice!

What would have happened?

Instructions: Complete the sentences by writing its corresponding word. Use capital letter where necessary. Use: hadn't looked / wouldn't have fixed / wouldn't have / wouldn't have realized, wouldn't have addressed /

1. If we _____ the defects for this project, we _____ them on time
2. If we hadn't tracked the defects for this project, we _____ them at all.
3. Most of the defects _____ if they _____ ?
4. If we _____ over these patterns, we wouldn't have realized that we needed to improve the requirements.
5. If we hadn't compared the defects found in testing with those found in production, we _____ how many defects our testing processes were not detecting.

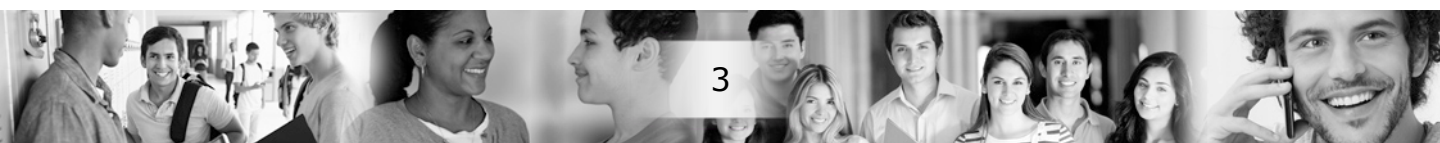


Let's Practice!

Would have been different?

Instructions: Read the statement. Select all the correct answers.

1. If defect tracking hadn't existed,
 - a. we had finished it first.
 - b. the personnel had a different perspective.
 - c. the project will have been different.
 - d. we couldn't have detected the problems in the software.
 - e. we couldn't have improved our software.





Extension Activity

Defect tracking in real life.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Find some way of tracking defects in real life.
- Write a 200 words text describing how the defects are tracked.
- Share with your partners.

Self-Assessment

Yes No

- I identify different ways of tracking defects.
- I produce a short text about defects tracking.
- I talk about tracking defects.

☐	☐
☐	☐
☐	☐



Target Words

- track** (verb): to follow and be attentive to how something evolves.
e.g. If we hadn't tracked the defects for this project, we wouldn't have addressed them on time.
- model for testing** (noun): an already planned process with specific indications for testing software.
e.g. Yet another advantage of defect tracking is the improvement in models for testing.
- utterly** (adverb): in a complete or absolute manner.
e.g. That's why it's utterly important to track defects, this is to ensure that there's clear information about how the defect should be treated.
- get acquainted** (expression): to become familiar.
e.g. I'm still getting acquainted with software development terms and I'm not sure what defect tracking is for.
- indispensable** (adjective): something that is necessary or essential.
e.g. As far as I'm concerned, many organizations consider this to be an indispensable process for the development of their projects.
- undeniable** (adjective): something that cannot be denied, something that is true.
e.g. On the course of a project, many defects are going to occur as an undeniable human error.



Learning Points

If Clauses with Past Perfect Form of Verb

We use the **third conditional** to talk about what the consequences of actions that did not occur could have been.

We use the **third conditional** when we want to explain or point out the possible consequences of a past event that did not take place.

Different from the *zero, first and second conditionals*, which refer to present or future events, the **third conditional** talks about unreal past events or actions.

The structure of the **third conditional** is the following:

If clause:

If + subject + past perfect of main verb.

Main clause:

subject + would/could/might + present perfect of main verb

Example:

*If we **had tracked** the defects before, the project **wouldn't have** been delayed.*

Note that if we invert the order of the sentences, we will not write a comma separating both sentences.

Example:

*The project **wouldn't have** been delayed if **we had tracked** the defects before.*

Examples

- *If we hadn't tracked the defects for this project, we wouldn't have addressed them on time or, even worse, we wouldn't have fixed them at all.*
- *So, most of the defects wouldn't have been fixed if they hadn't been tracked?*
- *if we hadn't looked at these patterns, we wouldn't have realized that we needed to improve the requirements' definition and coding areas.*
- *If we hadn't compared the defects found in the testing with those found in production, we wouldn't have realized how many defects our testing processes we are not detecting.*



Learning Points

Understanding the Concept of Defect Tracking

Defect tracking is the process in which the defects found in the program are tracked throughout the development of the project.

During the course of the development of a software, there are many defects or problems that are going to appear as a result of human errors. These defects can appear for various reasons and they can be encountered from the very beginning of the project, up to the last stages.

As throughout the project the team is going to find various defects, it is very important to track them properly. This means that the defects have to be entered in a database with a clear description of what the defect is about, to whom it is going to be sent, how severe it is, etc. This process is very important because sometimes the defects are going to have to be addressed by different people in order to validate that it is effectively a defect and not only a user's error. Therefore, whoever has to analyze the defect needs to have enough information available.

To deliver a project on time, the team needs to be very organized and make sure that the defects are fixed promptly. This is why, tracking defects will help the person in charge of fixing the defect to learn about the problem as soon as possible.

